

Juhyun (Julie) Lee

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Research Interests

Deep Learning: Multimodal Learning (Vision Language Model), Self-Supervised Learning, Agentic AI
AI for healthcare: Computational Pathology, Medical Image Analysis

Education

University of North Carolina at Chapel Hill

Master's in Computer Science

NC, United States
Aug. 2024 – May. 2026

Chung-Ang University

B.S. in Applied Statistics & B.A. in French Literature

Seoul, South Korea
Mar. 2020 – Aug. 2023

Publications

Structural MRI-Informed Multimodal Fusion for Robust Alzheimer's Disease Prediction

Juhyun Lee, Sungjoon Park, Kyungwook Lee, Soorin Yim, Doyeong Hwang, Dongyun Kim, Kiwoong Yoo, Soonyoung Lee, Kiyoung Kim
NeurIPS Imageomics Workshop, 2025

Proformer: a hybrid macaron transformer model predicts expression values from promoter sequences

Il-Youp Kwak, Byeong-Chan Kim, **Juhyun Lee**, Taein Kang, Daniel J Garry, Jianyi Zhang, Wuming Gong
BMC Informatics, 2024

Work Experiences

LinkedIn

AI/ML Engineer Intern

Mountain View, CA
May. 2026 – Aug. 2026

- Proposed a statistically rigorous AI agent evaluation framework and visualized its metrics on a dashboard, making quality regressions measurable and gating risky model/prompt changes.
- Instrumented OpenTelemetry spans capturing LLM tool calls and reasoning per turn, enabling end-to-end tracing that improved agent observability and sped up debugging.

LG AI Research

Research Scientist Intern (Mentor: Sungjoon Park, Kiyoung Kim)

Seoul, South Korea
May. 2025 – Aug. 2025

- Engineered a multimodal data processing pipeline and integrated it into a multimodal model, boosting Alzheimer's Disease classification accuracy by 20% over SOTA.
- Architected a modular Plan-Act-Memory AI agent to automate biomedical tasks using LangChain.

Research Experiences

UNC Carmichael Lab

Research Assistant (Advisor: Iain Carmichael)

NC, United States
Jan. 2026 – May. 2026

- Achieved 0.7 AUC in smoking status prediction from bladder cancer whole slide images by leveraging pathology foundation models and weakly supervised learning.

UNC ViVE Lab

Research Assistant (Advisor: Saif Khairat)

NC, United States
Aug. 2025 – Dec. 2025

- Fine-tuned Llama-3-8B using LoRA (PEFT) for clinical reasoning; deployed with vLLM and Flask.
- Developed a speech-to-speech system for real-time interaction using OpenAI API, WebRTC, and TypeScript.

UNC Yap Lab

Research Assistant (Advisor: Pew-Thian Yap)

NC, United States

Jan. 2025 – May. 2025

- Developed an Implicit Neural Representation (INR) framework that yielded 10x data compression while maintaining 92% reconstruction fidelity.

UNC Data Driven EnviroLab

Research Assistant (Advisor: Angel Hsu)

NC, United States

Aug. 2024 – Dec. 2024

- Implemented a RAG chatbot using LangChain, FAISS, and LLM APIs to deliver accurate, relevant responses on climate policy.

Chung-Ang University

Research Intern (Advisor: Il-young Kwak)

Seoul, South Korea

Mar. 2022 – June. 2023

Conference Presentations

- **UNC Innovate Cancer Data Science Symposium** Jan. 2026
Poster Presentation
- **NeurIPS Imageomics Workshop** Dec. 2025
Poster Presentation
- **UNC Graduate Research Symposium** Oct. 2025
Poster Presentation
- **UNC Data Science Day** Sep. 2025
Poster Presentation
- **International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting** May 2025
Oral Presentation

Academic Services

Reviewer

ICML 2026, NeurIPS 2026

Mentorship

UNC Graduate Women in Computer Science Mentor

Teaching

- Instructor for *Medical Image Analysis and Computer Vision* for UNC PRIMES Summer School 2025

Awards/Honors

- Summa Cum Laude**, Chung-Ang University Aug. 2023
- Random Promoter Dream Challenge 2022 (3rd place)** Sep. 2022
- Department Honor Scholarship**, Chung-Ang University Aug. 2020

Skills

Programming Languages: Python, MATLAB, Java, JavaScript, SQL, R

ML & Data Science: Pytorch, TensorFlow, LangChain, Scikit-Learn, NumPy, Pandas, Matplotlib, SAS

Web Development: SpringBoot, Git, VueJS, AWS